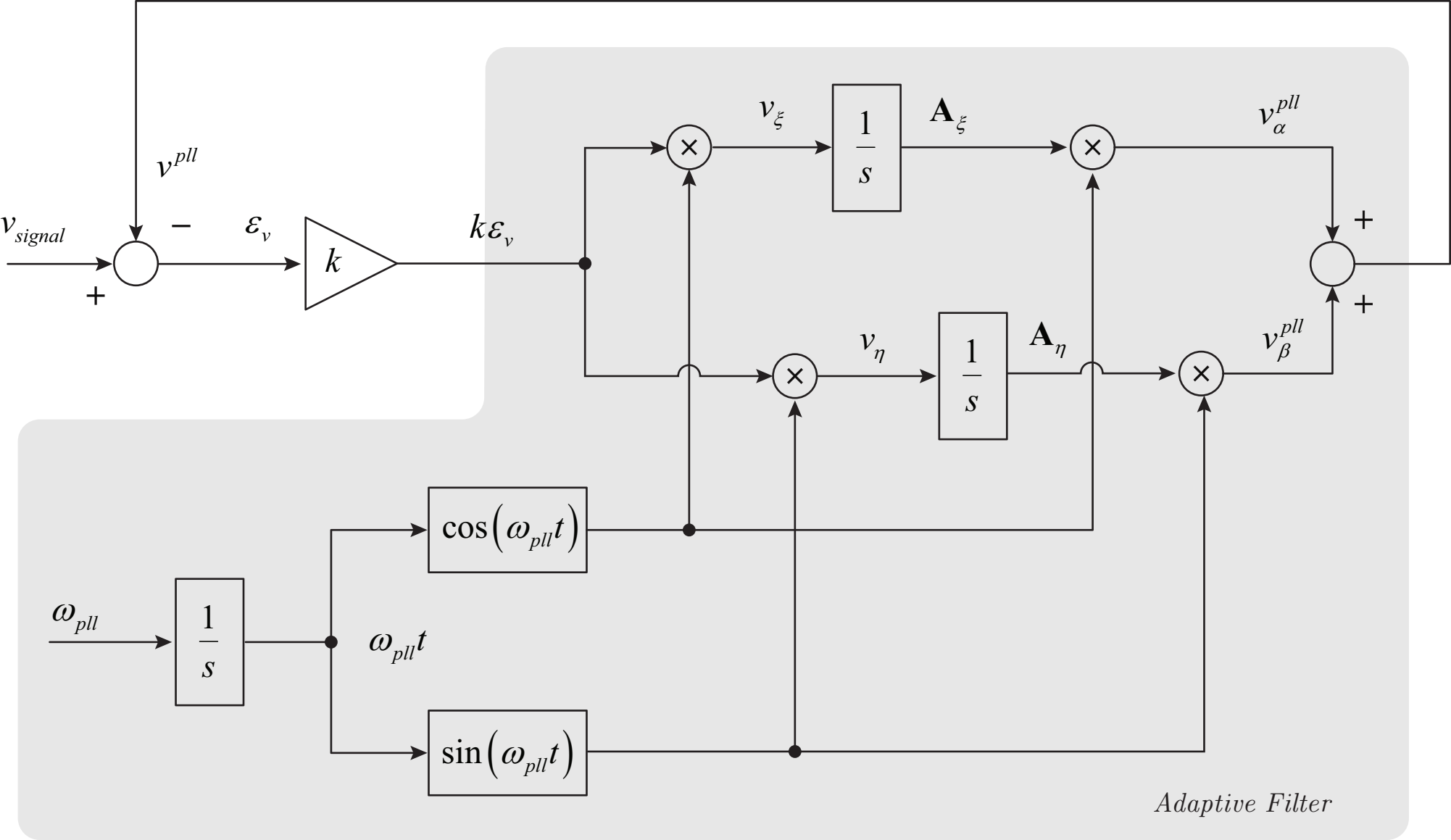
v^{pll}
v_{signal}
$-$
$+$
ε_v
k
$k\varepsilon_v$
v_ξ
$\frac{1}{s}$
$\mathbf{A}_\xi$
v_α^{pll}
v_η
$\frac{1}{s}$
$\mathbf{A}_\eta$
v_β^{pll}
$+$
$+$
ω_{pll}
$\frac{1}{s}$
$\omega_{pll}t$
$\cos(\omega_{pll}t)$
$\sin(\omega_{pll}t)$
Adaptive Filter